



POLITECNICO
MILANO 1863

SCUOLA DI INGEGNERIA INDUSTRIALE
E DELL'INFORMAZIONE

MSC THESIS DEFENCE · AERONAUTICAL ENGINEERING

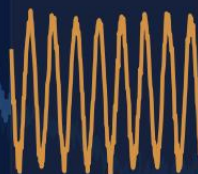
Generative Digital Twin Methods for Spacecraft Telemetry Anomaly Detection

Toward Data-Driven Fault Detection, Isolation and Recovery (FDIR) Design and Testing

Candidate **Mehrab Jamshidi** · ID 10976258

Advisor **Prof. Andrea Colagrossi**

Politecnico di Milano · Academic Year 2025–26 · Defence 22 July 2026



Spacecraft FDIR has a structural blind spot

POINT ANOMALY

The signal leaves its amplitude range.

A spike, step, or hard-over that crosses a limit.

Threshold logic: caught

CONTEXTUAL ANOMALY

The rhythm or correlation breaks —
amplitude stays perfectly in bounds.

A slow drift, a phase slip, a broken periodicity.

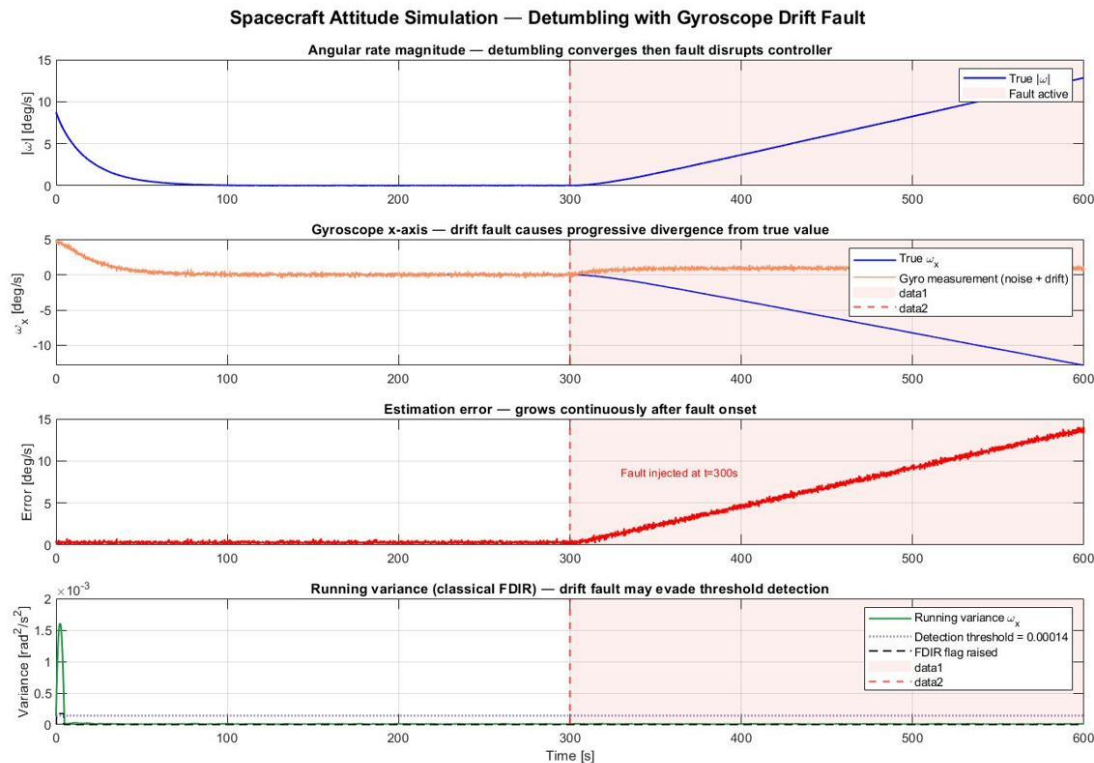
Threshold logic: invisible

43 of 105

real in-orbit anomalies are contextual (NASA SMAP / MSL benchmark).

No threshold tuning fixes this — there is no amplitude excursion to trip on.

A controlled simulation exposes the gap



1 Closed-loop attitude dynamics

Simulink · PD detumbling with a gyroscope model.

2 Gyro drift injected at $t = 300$ s

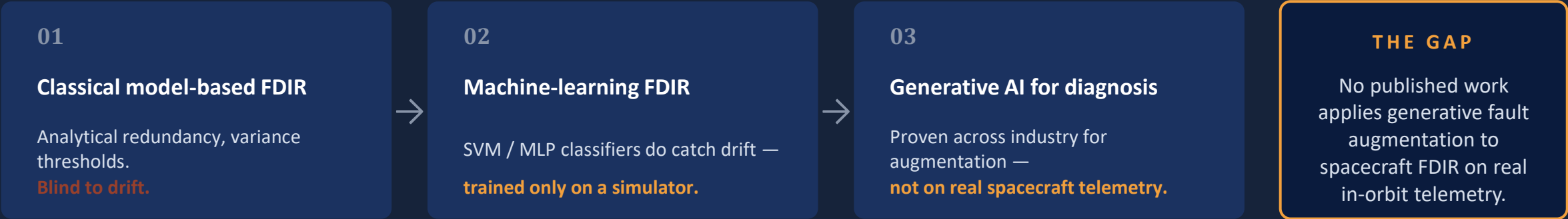
estimation error grows to ≈ 14 deg/s.

3 The FDIR flag is never raised

running variance never crosses its threshold.

Structural, not a tuning artefact: a slow drift adds a near-constant offset inside any short window, so the variance barely moves.

The data-driven turn — and its untested assumption



CLOSEST PRECEDENT

	Crotti & Colagrossi (2025)	This work
Faults	simulator, parametric models	real in-orbit telemetry
Detectors	supervised classifiers	label-free + supervised
Fault generator	—	learned, controllable

Real in-orbit telemetry makes the study possible

PRIMARY DATASET

NASA SMAP / MSL

Hundman et al. (2018) — expert-labelled satellite & rover telemetry.

- ▶ 82 channels (81 unique) · 55 SMAP / 27 MSL
- ▶ 105 expert-verified anomalies — 62 point · 43 contextual
- ▶ Clean train/test split · point-adjust · a published result to reproduce

CONSIDERED, NOT USED

ESA Anomaly Benchmark

Kotowski et al. (2024) — 3 missions, > 700 M points each, multivariate. Beyond this scope, but flagged as the natural next dataset.

0.39%

minimum per-channel anomaly rate

48%

of channels have < 5% anomalies

12.28%

mean per-channel anomaly rate

The compounding difficulty: anomalies are rare AND often invisible to amplitude thresholds.

Can generative models trained on real NASA SMAP/MSL telemetry improve fault detection over classical FDIR — **and how far can synthetic fault data substitute for, augment, and characterise** the detectors an operational FDIR system would deploy?

01

PHASE 2

Detectors through the FDIR lens

VAE + LSTM benchmarked through the point / contextual taxonomy; the pipeline validated against the published benchmark.

02

PHASE 3

A WGAN-GP on real fault telemetry

To my knowledge the first conditional GAN trained on real spacecraft faults — with four architectural contributions.

03

PHASE 4A

Clean-protocol operational value

Where synthetic data actually helps a deployed detector, established under a leakage-free held-out split.

04

PHASE 4B

Sensitivity → FDIR design data sheet

F1 and latency mapped across fault severity and duration, on channels the generator never saw.



★ **Contribution 4 is starred because it is the principal, most novel result** — an FDIR design data sheet from a controllable fault generator. The rest of the talk builds toward it (Phase 4B).

One question, five phases

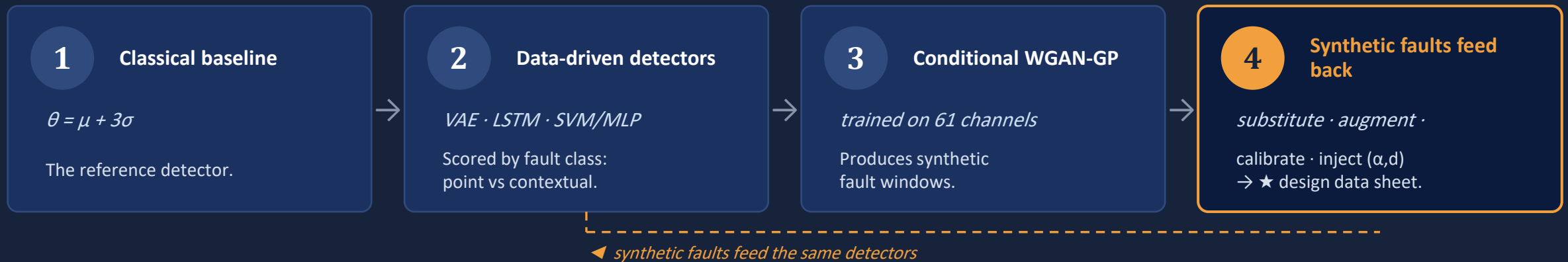
detection first, then generation, then generation used as a design instrument.



The destination is Phase 4B: a controllable generator turned into a queryable design surface for FDIR.

One system, and the two rules behind every number

INPUT Real telemetry — 81 channels (nominal + labelled faults)



TWO RULES THAT GOVERN EVERY NUMBER IN THIS TALK

1 Two thresholds

oracle — best F1 using test labels (an upper bound).

k = 3 deployable — $\theta = \mu + 3\sigma$ on prediction error, no labels.

2 Clean held-out protocol

generator trained on 61 channels, evaluated only on 20 it never saw · paired Wilcoxon, $n = 20$.

An earlier contaminated run is reported and withdrawn.

Data-driven detectors close the contextual gap

Method	Contextual (n = 30)	Point (n = 51)	Labelled faults?
Z-score threshold (baseline)	0.052	0.259	no
VAE — k = 3 (deployable)	0.319	0.474	no
VAE — oracle (upper bound)	0.638	0.561	no
SVM — RBF (supervised) ‡	0.508	0.651	yes
LSTM — k = 3 (deployable)	0.717	0.628	no
LSTM — oracle (upper bound)	0.836	0.688	no

On the contextual class classical FDIR cannot reach:

0.052 → **0.717**

same fixed 3σ rule · no labelled faults · LSTM deployable

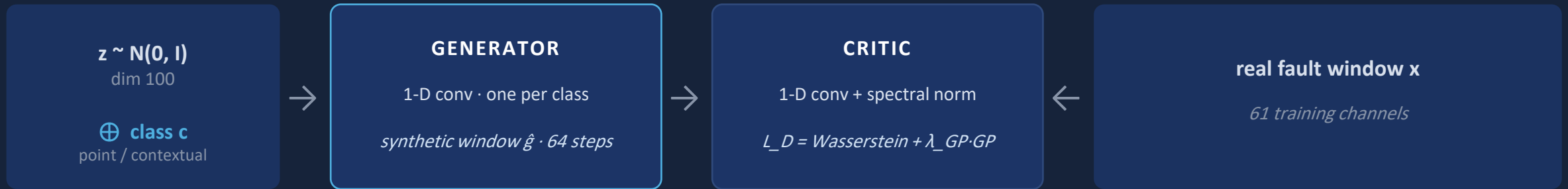
WHY THE LSTM?

It predicts how the signal should evolve — contextual faults violate those causal dynamics. The VAE only reconstructs a window, so it is strong but order-blind.

Threshold / VAE / LSTM: point-adjust on the full split — like-for-like. ‡ SVM scored at window level, a different protocol (B8). **Oracle rows use test labels — an upper bound, not an operating point.**

A conditional WGAN-GP for fault synthesis

Why generate? A supervised detector needs labelled faults a spacecraft may not have before its first in-orbit anomaly.



★ **ACF-matching loss** $\lambda_{ACF} \cdot (1/K) \sum_k (\rho_k[\hat{g}] - \rho_k[x])^2$ over lags 1–20 — direct supervision of temporal structure, back-propagated into the generator.

FOUR ARCHITECTURAL CONTRIBUTIONS BEYOND THE STANDARD WGAN-GP

01

1-D convolutional generator & critic

local temporal patterns

02

Autocorrelation-matching loss ★

direct supervision of structure

03

Per-class specialised generators

no cross-class mode-averaging

04

Spectral normalization in the critic

gradient-penalty compatible

$\lambda_{GP} = 10 \cdot \lambda_{ACF} = 5 \cdot K = 20 \text{ lags} \cdot n_{critic} = 5 \cdot \text{latent dim } 100 \cdot \text{seed } 42$

Synthetic faults reproduce the real distribution



t-SNE — synthetic contextual (light) overlaps the real distribution (green).

Metric	Contextual	Point
Fréchet distance (6-feature)	0.141	0.403
ACF mean $ \Delta $ (lags 1–20) smaller is better · amber = the better of the two classes	0.026	0.015

Contextual reproduced best on Fréchet.

A consistent oscillatory structure the convolutional generator captures well.

Point faults are heterogeneous.

Spikes, steps, hard-overs — one generator covers a wider distribution. Heterogeneity, not a training failure.

Both classes match the real autocorrelation within 0.03. Honest caveat: no controlled ablation — evidence, not proof.

Synthetic transfers as well as real faults from another channel

synthetic only

generator never saw the channel



cross-channel real

faults borrowed from the 61 training channels



the channel's own faults †

Phase-2 leave-one-out ceiling



MATCHED PROTOCOL

synthetic $\approx$ cross-channel real $\rightarrow$ **p = 0.814**

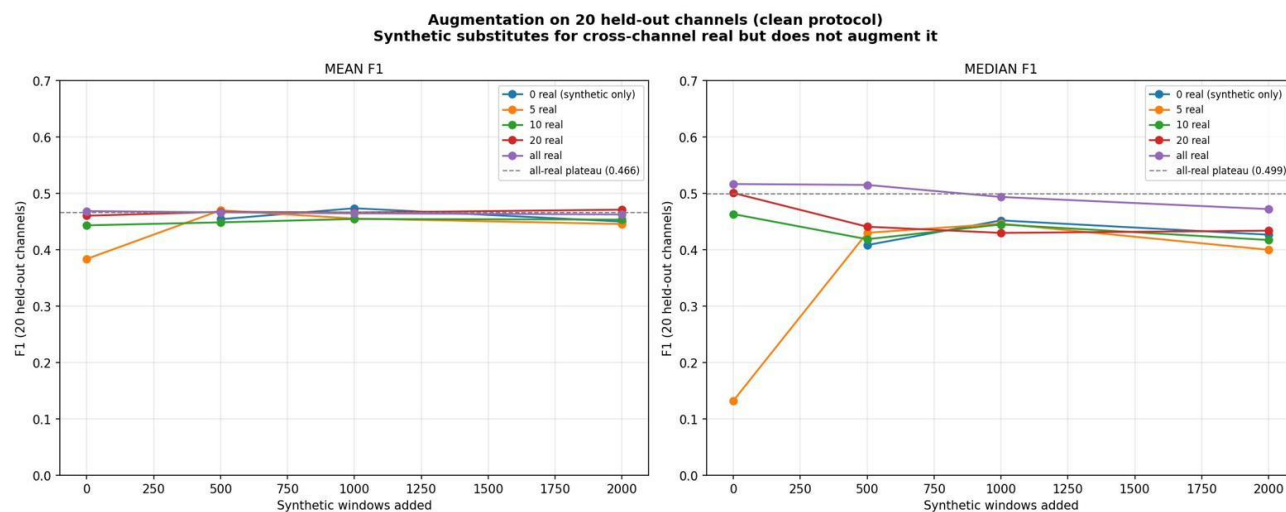
Synthetic and cross-channel real are statistically indistinguishable. Neither source is privileged.

By class: 0.57 on 14 point channels, 0.24 on 6 contextual — this tracks the SVM's own weakness on contextual faults, not the generator's quality.

† The 0.631 own-channel bar uses a different protocol (own faults, leave-one-out) in both training data and evaluation set — a reference ceiling, not a like-for-like comparator. **The matched pair is 0.474 vs 0.469 (see B8).**

mean F1 · all evaluated on the 20 held-out channels the generator never saw

What synthetic data is — and is not — worth



Clean-protocol grid — **mean F1** (left) and **median F1** (right) across real $\times$ synthetic windows.



Substitution works

Synthetic-only retains $\approx 75\%$, no detectable loss ($p = 0.13$).
The real deployment case.



Augmentation does not help

Mean stays flat, **median drops** as synthetic is added — see the grid.



Calibration does not beat $k = 3$

Both plateau ≈ 0.47 ; classical 3σ stays the default.

Why the negatives? They depend on the nominal score distribution — not the fault distribution generative modelling improves. So synthetic data substitutes for missing rare faults; it does not augment an adequate dataset or calibrate a threshold. The negatives sharpen the claim.

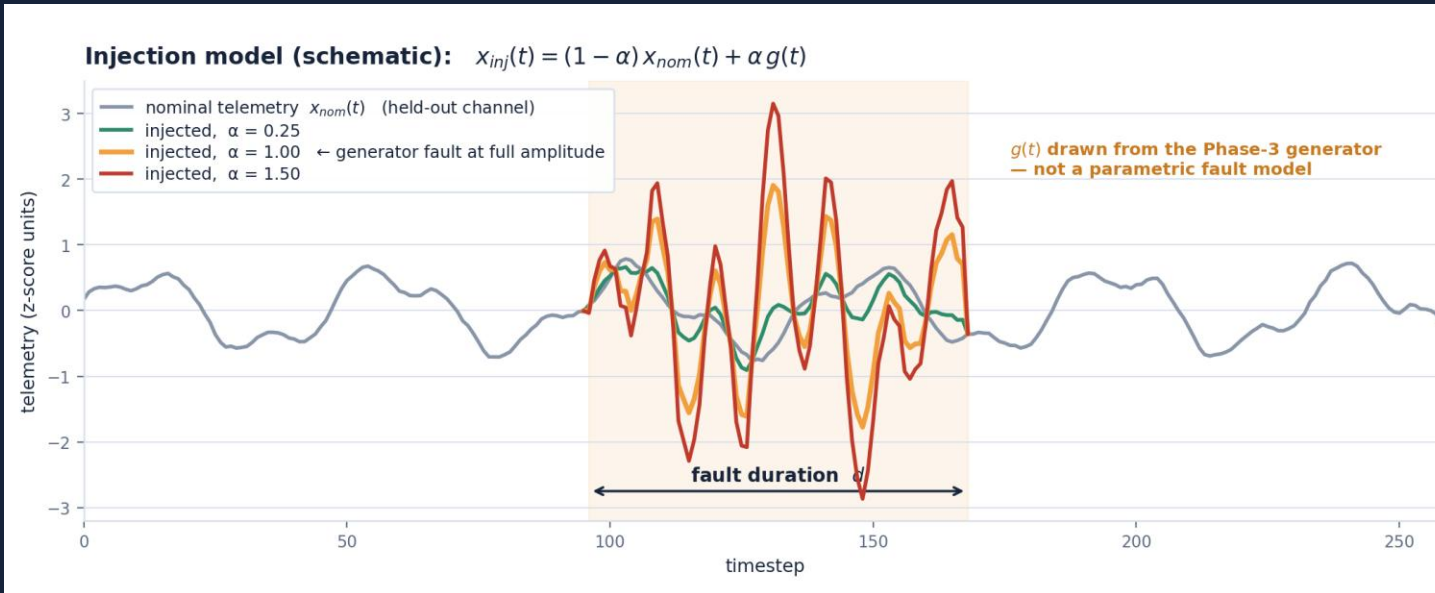
How the design surface is built

THE INJECTION MODEL

$$x_{inj}(t) = (1 - \alpha) \cdot x_{nom}(t) + \alpha \cdot g(t)$$

α severity 0.10–1.50 · d duration 16–256 steps

$g(t)$ — a synthetic fault window from the generator, not a parametric formula.



EXPERIMENTAL DESIGN

6 severities × 5 durations × 5 replicates

≈ 9,000 injections · held-out channels only

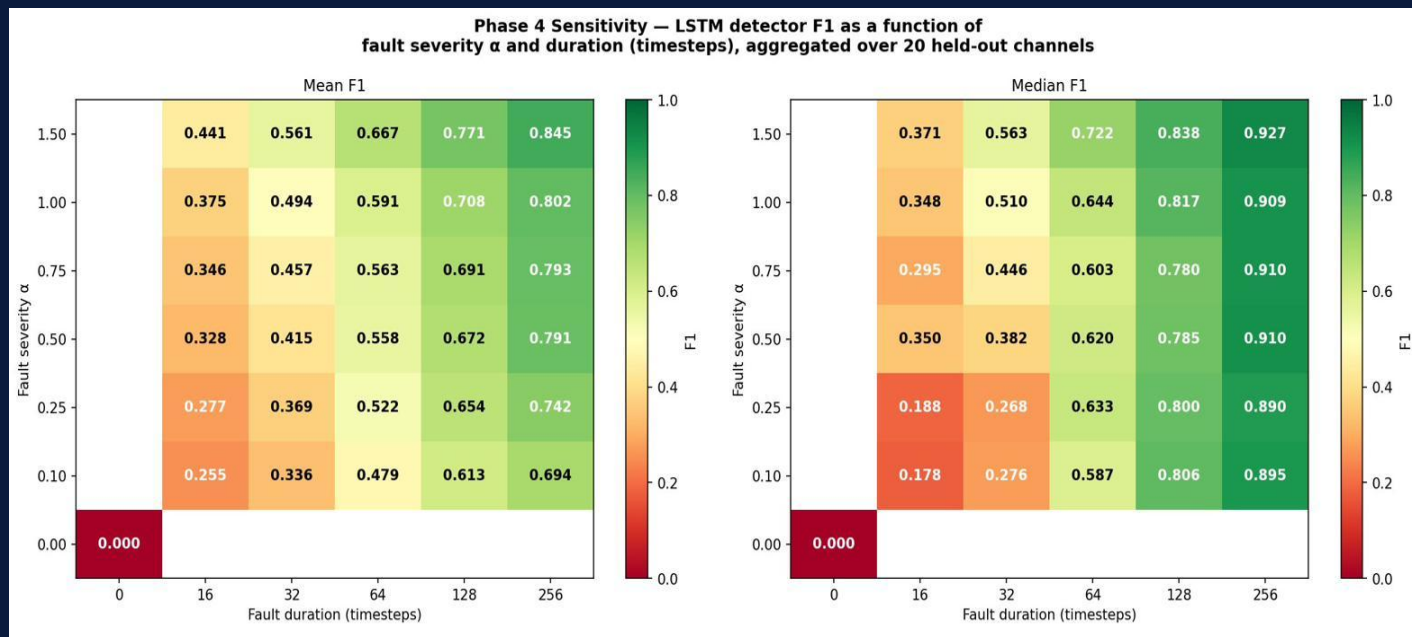
Detector: LSTM at deployable $k = 3$.

✓ THE VALIDITY CONTROL

No-injection false-positive baseline: **median FP 0.011.**

The surface is not contaminated by over-flagging (mean 0.123, inflated by 2 of 20 channels — B3).

An FDIR design data sheet from a controllable fault generator



Mean detector F1 across severity α (rows) and duration d (columns), on 20 held-out channels.

THE IDEA

Inject faults at controlled severity α and duration d — then map detector F1 as a 2-D surface, on channels the generator never saw.

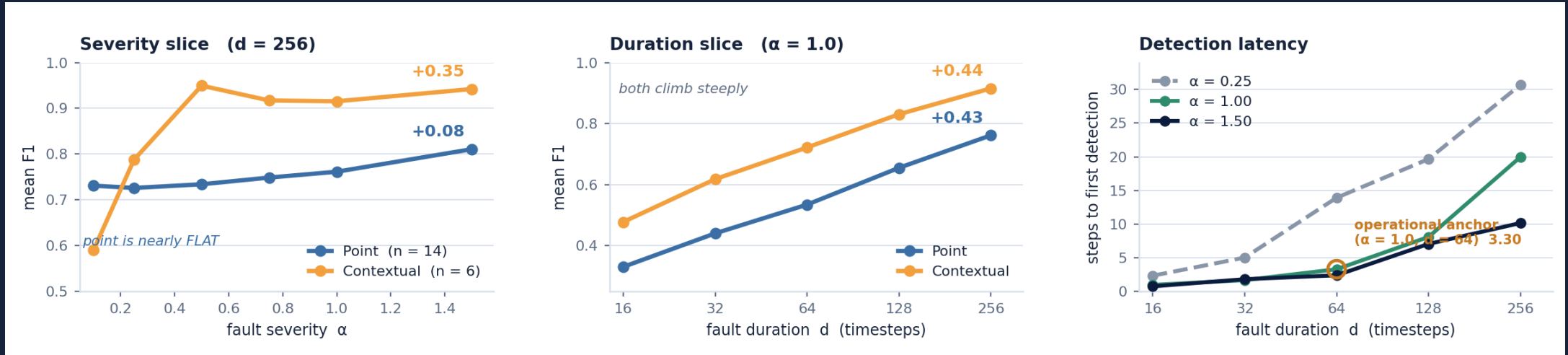
F1 SPANS

0.255 → **0.845**

smallest / shortest → largest / longest · the mean surface rises monotonically in both axes.

No fixed labelled dataset can produce this: each real fault occurs at one uncontrollable magnitude and duration. A controllable generator turns detection capability into a queryable design surface.

Reading the design data sheet



1 Severity matters only for contextual

At $d = 256$, $\alpha 0.10 \rightarrow 1.50$ moves point channels by only $+0.08$ — but contextual by $+0.35$ ($0.590 \rightarrow 0.942$).

2 In aggregate, duration dominates

At $\alpha = 1.0$, duration $16 \rightarrow 256$ gains $+0.43$ — for both classes. A fault must persist to be seen.

3 Latency is a design constraint

Severe + short: ≈ 1 step. Subtle + long: ≈ 27 . Anchor ($\alpha = 1.0, d = 64$): F1 0.591, latency 3.30.

Honest scope: this characterises this detector under the point-adjust protocol — it is not a universal property of the fault classes. P-1, an unseen channel: LSTM F1 = 0.718 on its real anomalies, against 0.011 for the baseline.

Generative models improve FDIR — in two roles that must not be conflated

ROLE 1

As detectors

The VAE and LSTM close the contextual gap classical FDIR cannot reach — the LSTM most strongly.

Contextual class: **0.836 oracle** · **0.717 deployable**.

No labelled faults required · validated to Δ 0.006.

ROLE 2 ★

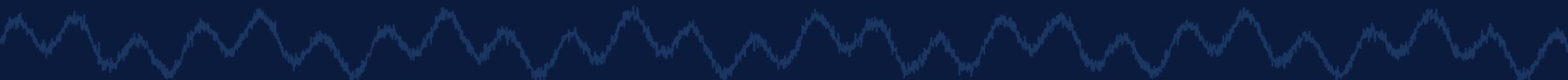
As a design instrument

A controllable fault generator characterises what a detector can and cannot see across the operational envelope.

On channels it never saw.

A capability real data fundamentally lacks — the clearest value emerged here, not as a competitor to real data.

The clearest value emerged not as a competitor to real data, but in a capability real data cannot provide.



A narrow — and therefore trustworthy — recommendation

✓ Deploy a fault generator to...

- train a supervised detector for fault types too rare to have real examples;
- act as a controllable testbed to characterise detection logic before flight.

✗ Do not deploy it to...

- augment an already-adequate real dataset — it substitutes, it does not augment;
- calibrate an unsupervised threshold — classical $k = 3$ remains the default.

LIMITATIONS

- Univariate — context channels unexploited.
- $n = 20$ limits statistical power.
- The surface characterises generator faults under point-adjust.
- The four GAN contributions were adopted together — no ablation.

FUTURE DEVELOPMENTS

- **Multivariate detection & generation — highest priority.**
- Scale to the ESA benchmark; stronger architectures.
- Label-free threshold calibration.
- Validate the data sheet against hardware-in-the-loop → a Cognitive (L5) Digital Twin.



POLITECNICO
MILANO 1863

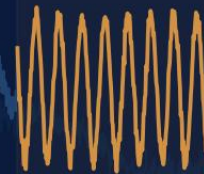
Thank you.

Questions welcome.

Data-driven detectors close the contextual-anomaly gap left open by classical FDIR — and a controllable generative model of faults becomes a design instrument for characterising detection capability before flight.

Mehrab Jamshidi · Advisor: Prof. Andrea Colagrossi

Politecnico di Milano · MSc Aeronautical Engineering · 2025–26



BACKUP

Supporting material for questions

[B1](#) [Full method comparison — all 81 channels](#)



[B2](#) [WGAN-GP architecture, training & the limits of the evidence](#)



[B3](#) [Sensitivity — injection model & robustness](#)



[B4](#) [Dataset statistics & the held-out split](#)



[B5](#) [Point-adjust — the protocol and its two caveats](#)



[B6](#) [Results by fault class — transfer & sensitivity](#)



[B7](#) [Per-channel detector distributions](#)



[B8](#) [Evaluation protocols — what is scored how](#)



Full method comparison — all 81 channels

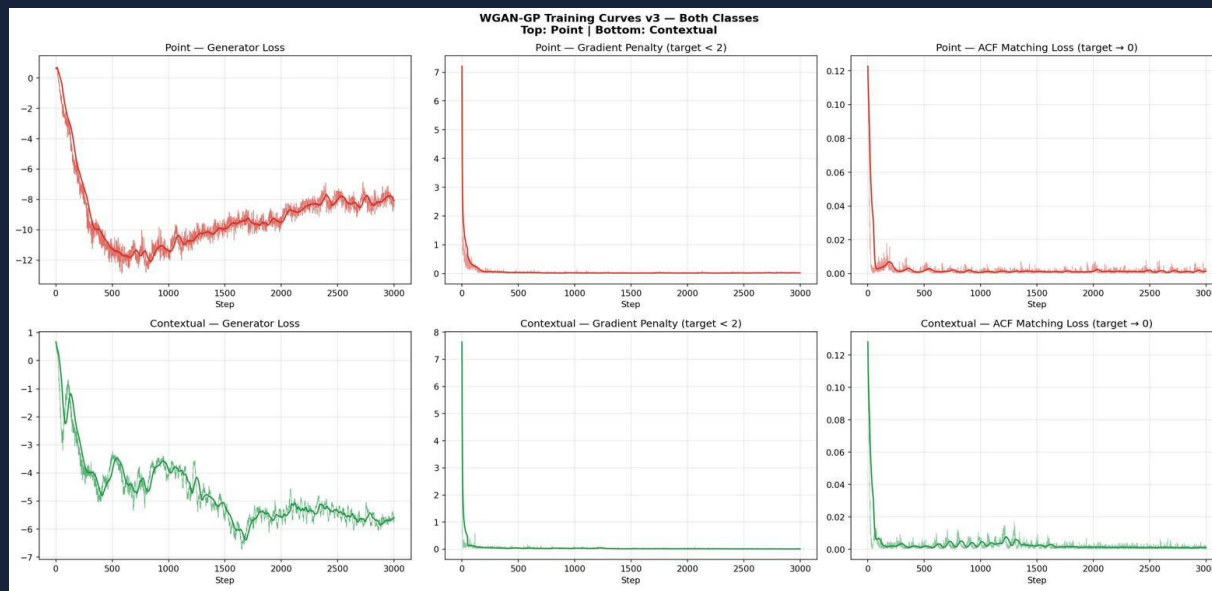
Method	Mean F1	Median F1	F1 > 0	F1 > 0.5
Z-score threshold (baseline)	0.183	0.019	46 / 81	12 / 81
ANN classifier (MLP 32→16)	0.413	0.253	70 / 81	32 / 81
VAE — point-adjust (k = 3)	0.417	0.327	58 / 81	37 / 81
VAE — point-adjust (oracle)	0.590	0.674	76 / 81	52 / 81
SVM classifier (RBF kernel)	0.598	0.690	81 / 81	46 / 81
LSTM regression (k = 3)	0.661	0.849	78 / 81	55 / 81
LSTM regression (oracle)	0.743	0.918	81 / 81	63 / 81

Ordered by mean F1. Threshold, VAE and LSTM are scored per time step under point-adjust on the full test split and are strictly like-for-like; the SVM and ANN are evaluated at window level, their natural unit of prediction — a different protocol (see B8). Published Hundman (2018) LSTM: SMAP 0.752, MSL 0.696 — reproduced here at 0.746 / 0.738.

Conditional WGAN-GP — architecture, training, evidence

Hyperparameter	Value
Latent dimension	100
Generator steps	3,000
Critic steps / gen (n_critic)	5
Gradient-penalty λ_{GP}	10
ACF-loss weight λ_{ACF}	5.0
Gen / critic learning rate	1e-4 / 4e-4
Adam (β_1, β_2)	(0.0, 0.9)
Batch size	64
Random seed	42

Point generator: 2,000 real point windows · Contextual: 792 (61 training channels).
 Generator 184,865 params · Critic 42,337 — size is not the onboard-inference bottleneck.



▲ WHAT THE EVIDENCE DOES NOT ESTABLISH

No ablation. The four contributions were adopted together; their individual marginal effect is not isolated. The ACF loss decays to ~0 partly by construction.

No memorisation test. Indirect evidence against pure memorisation: the point generator produces windows measurably smoother than the sharpest real spikes — which is why its Fréchet (0.403) is the weaker of the two.

TRAINING OBJECTIVE

$$\text{Critic } L_D = E[D(G(z|c))] - E[D(x|c)] + \lambda_{GP} \cdot GP$$

$$\text{Generator } L_G = -E[D(G(z|c))] + \lambda_{ACF} \cdot (1/K) \sum (\rho_k[G(z|c)] - \rho_k[x|c])^2$$

Sensitivity analysis — injection model & robustness

Injection model

$x_{inj} = (1 - \alpha) \cdot x_{nom} + \alpha \cdot g$, with fault window g drawn from the Phase 3 generator; α = severity, d = duration.

α semantics

$\alpha = 1 \rightarrow$ the synthetic fault at its intended amplitude (nominal fully replaced); $\alpha < 1 \rightarrow$ attenuated; $\alpha > 1 \rightarrow$ an affine extrapolation, $x_{nom} + \alpha(g - x_{nom})$.

Scaling

Generator output is in tanh space $[-1, 1]$; it is multiplied by 3 into z-score units, so $\alpha = 1$ means the same thing on every channel.

Long faults

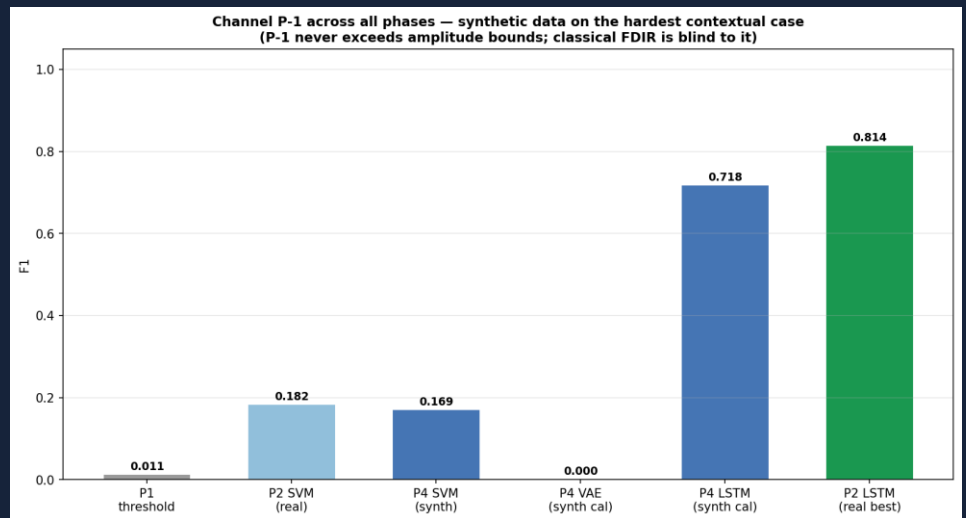
The generator window is 64 steps, so $d = 128$ and $d = 256$ are built by concatenating generator samples — a construction limit of the current window length.

Validity control

No-injection false-positive baseline: median FP rate 0.011 — the surface is not contaminated by over-flagging. (Mean 0.123, inflated by exactly 2 of 20 channels, D-5 and R-1, which saturate at 1.0.)

Cell membership

20 channels up to $d = 128$; 19 at $d = 256$ (one channel has no clean segment long enough). Latency is averaged only over DETECTED injections, so it must be read jointly with F1.



P-1 across all phases — the hardest contextual case, as a held-out unseen channel.

Dataset statistics & the held-out split

Property	Value
Total telemetry channels	82 (81 unique)
SMAP / MSL channels	55 / 27
Labelled anomaly windows	105
Point / contextual anomalies	62 / 43
Min anomaly rate (per channel)	0.39%
Max anomaly rate	50.07%
Mean / median rate	12.28% / 6.02%
Channels < 5% anomalies	39 of 82 (48%)
Channels with only 1 anomaly	63 of 81

CLEAN HELD-OUT SPLIT (seed 42)

61 training channels → generator training pool

20 held-out channels → evaluation only (never seen by the generator)

Composition: 13 point + 6 contextual + 1 mixed (C-1, labelled both). C-1 is grouped with the point class in every Phase-4 class-split, so the held-out set is reported as 14 point + 6 contextual. Slide 10's 81-channel split uses the opposite rule — any channel with a contextual anomaly counts as contextual (4 are mixed: C-1, C-2, G-7, T-1). Stratified by primary anomaly class (three strata), not by spacecraft: 16 SMAP / 4 MSL.

HELD-OUT CHANNELS (20)

A-8, B-1, C-1, D-4, D-5, D-9, D-14, E-8, E-10, F-3, G-1, G-2, G-3, G-4, M-2, P-1, P-2, R-1, T-2, T-5

P-1 — the illustrative contextual channel — falls in the held-out set, so the generator never saw it. (The detectors still learn its NOMINAL behaviour, which is unsupervised and fault-free.)

Point-adjust — the protocol and its two caveats

What it is. If any single step inside a labelled ground-truth segment is flagged, the whole segment is counted as detected. It encodes the operational goal — trigger at least once within a fault event — and is the standard on this benchmark.

CAVEAT 1 — ABSOLUTE INFLATION

Kim et al. (2022): point-adjust inflates F1 on long segments, so these scores are NOT comparable to point-wise metrics.

Why it is still correct here. It is the protocol Hundman and Xu report under, so it makes pipeline validation like-for-like; and it is applied identically to the threshold, VAE and LSTM, so the relative ordering is computed on one consistent basis.

CAVEAT 2 — CONDITIONS THE DURATION AXIS

One flagged step credits the whole d-step segment as a true positive, while the false-positive count is fixed by the clean stream — so precision rises with duration partly by construction.

At $\alpha = 1.0$: $d = 16 \rightarrow$ precision 0.313 / recall 0.773; $d = 256 \rightarrow$ precision 0.756 / recall 0.939. Precision carries +0.44 of the gain, recall +0.17.
Severity axis is NOT affected ($d = 256$: precision 0.661 $\rightarrow$ 0.791, recall 0.823 $\rightarrow$ 0.986).

How to read every F1 in this talk. The ORDERING of detectors and the SEVERITY axis are robust; only the MAGNITUDE of the duration axis is protocol-conditioned. Every F1 is benchmark-comparable under point-adjust — not point-wise. Supervised classifiers are scored at window level (§2.3.2); full protocol map on B8.

Results by fault class — transfer & sensitivity

SYNTHETIC → REAL TRANSFER (SVM)

Point (n = 14) **0.57** Contextual (n = 6) **0.24**

Weakness tracks the SVM, not the generator (see paradox, right).

SENSITIVITY EFFECTS BY CLASS

Severity (α 0.10→1.50, d = 256):

point +0.08 (0.731→0.811)

contextual +0.35 (0.590→0.942)

Duration (d 16→256, α = 1.0):

point +0.43 (0.331→0.762) contextual +0.44 (0.478→0.915)

THE APPARENT PARADOX

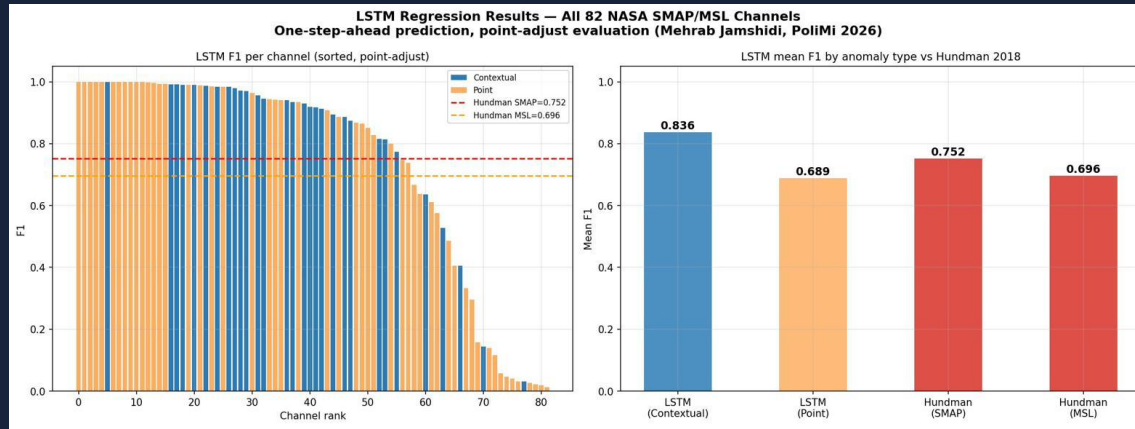
The generator reproduces contextual faults best (Fréchet 0.141), yet the synthetic-trained SVM scores worst on them. The SVM's 6 hand-features are amplitude-dominated — even real-trained, it reaches only 0.508 on contextual. The ceiling is the classifier, not the generator.

Design consequence. DURATION governs detectability for both classes; SEVERITY discriminates only contextual. Hardest cell: contextual α = 0.10, d = 16 → F1 0.168; sustained to d = 256 → 0.590.

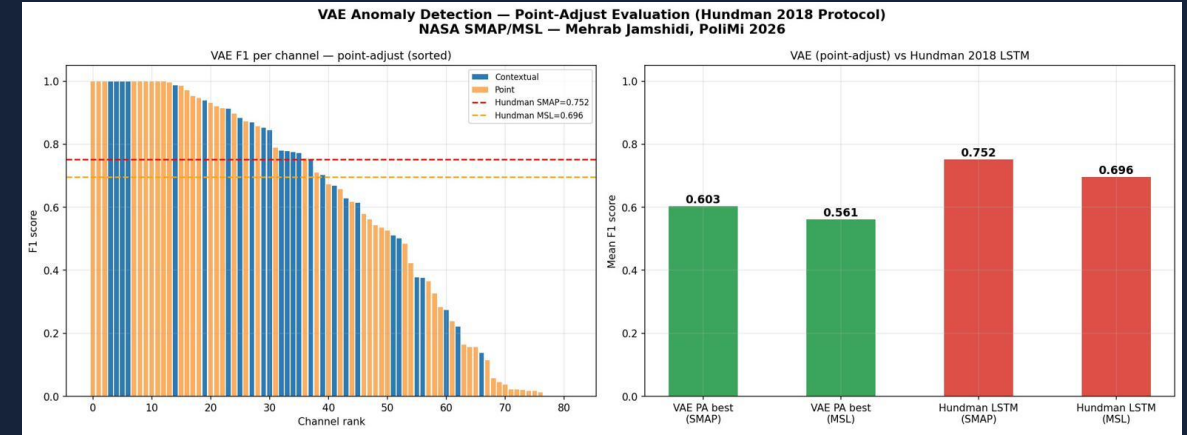
VAE by class: k=3 contextual 0.319 / point 0.474 vs oracle 0.638 / 0.561. The deployable gap is large because 26 of 81 channels peak at k = 1.0, which k = 3 cannot replicate — so the LSTM, not the VAE, is the recommended deployable detector.

Net: point faults are detectable once they persist; contextual faults need both persistence and amplitude. This is the design content of the sensitivity surface, resolved by class.

Per-channel detector distributions (all 81 channels)



LSTM — per-channel F1, sorted



VAE — per-channel F1, sorted

LSTM. Contextual mean 0.836, point 0.688 (oracle). Exceeds the Hundman SMAP line (0.752) on a large fraction of channels and beats the MSL line overall (0.738 vs 0.696).

VAE. 0.604 SMAP / 0.561 MSL at oracle — consistently below the LSTM on the contextual class.

Both are right-skewed. A minority of channels sit near zero and drag the mean down, which is why every table reports mean AND median.

The per-channel spread is the honest limitation of a univariate approach — a minority of channels remain hard for every detector, which is exactly what motivates the multivariate future work.

Evaluation protocols — what is scored how

Detector	Unit	Evaluation set	Protocol
Z-score threshold (P1)	time step	full test split	point-adjust
VAE reconstruction error (P2)	time step	full test split	point-adjust
LSTM prediction error (P2)	time step	full test split	point-adjust
SVM / ANN, real faults (P2)	window	1 anomalous vs 10 nominal, pooled over folds	leave-one-out CV
SVM synthetic / cross-channel real (P4)	window	full test split	single fit

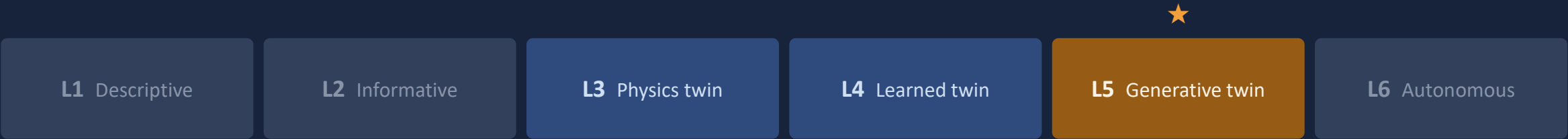
1. The three stream detectors (amber) are scored identically. The central claim of this thesis — contextual F1 from 0.052 to 0.836 oracle, 0.717 deployable — rests only on them, and is strictly like-for-like.

2. The supervised classifier is scored at window level (§2.3.2); its small per-fold sets are “indicative rather than definitive” (§3.6). It follows the precedent literature and no conclusion depends on it.

3. The Phase-4 pair is protocol-matched: synthetic (0.474) and cross-channel real (0.469) are both single fits on the full test split — which is why slide 14 reports p = 0.814, and why the 0.631 own-channel figure carries a dagger. Point-adjust and its two caveats: see B5.

The Digital Twin the title refers to

Maturity levels after Wei et al. — “Digital Twin” names a trajectory, not a single artefact.



Physics twin · L3

Classical FDIR

The Simulink attitude model — a physics-based digital twin of nominal behaviour.

Learned twin · → L4

VAE + LSTM

Learn nominal behaviour directly from real telemetry — a data-driven twin.

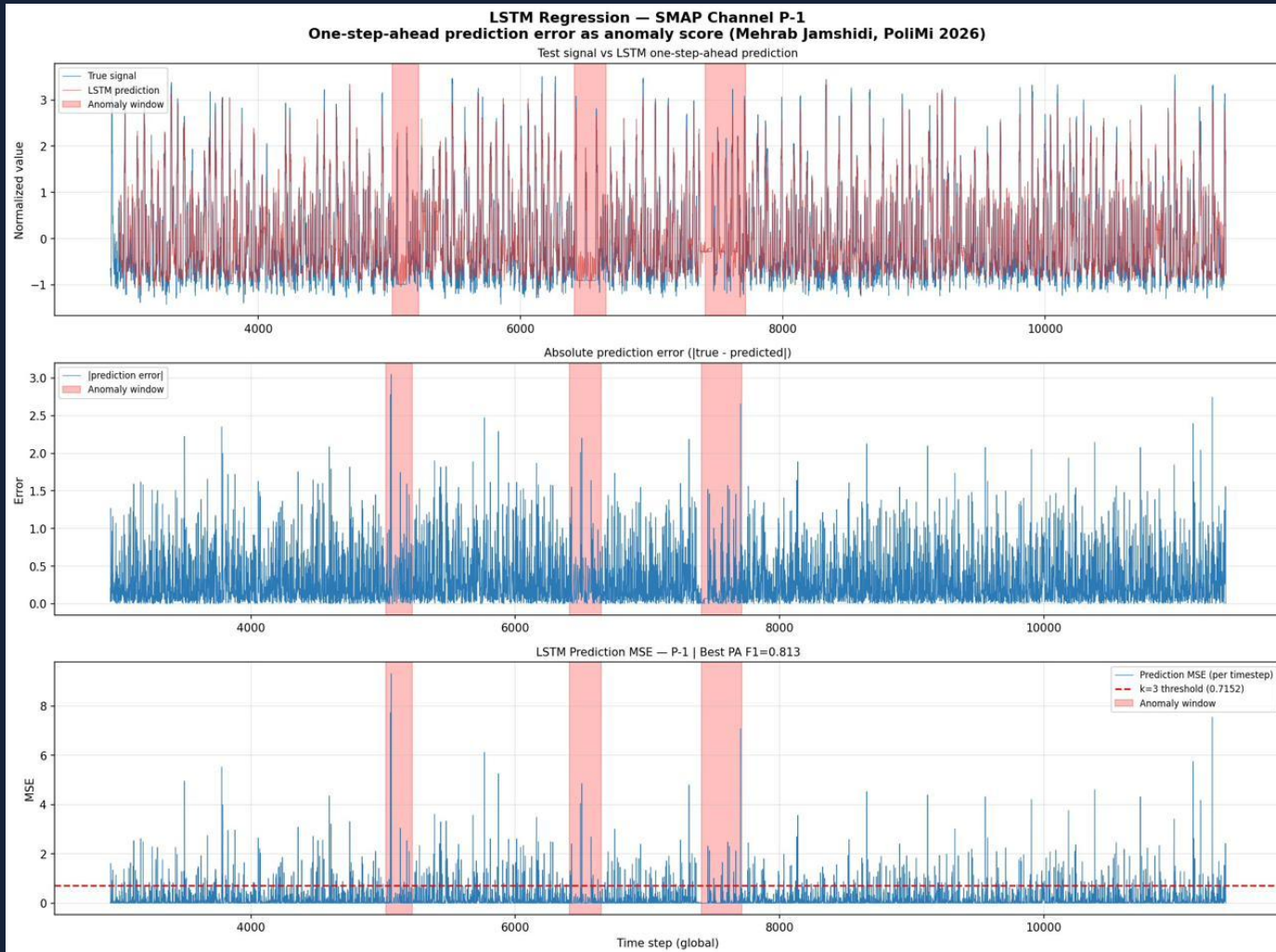
Generative twin · → L5 ★

WGAN-GP

A controllable model of the fault distribution — a twin that can characterise its own detection capability.

This work supplies the generative component that carries FDIR toward the Cognitive (L5) level.

The pipeline reproduces the published benchmark



REPRODUCTION · LSTM vs Hundman (2018)

SMAP **0.746** vs 0.752 published ($\Delta 0.006$)

MSL **0.738** vs 0.696 published ($+0.042$)

Reproduction to within 0.006 validates the pipeline.

P-1 — the hardest contextual channel

0.011

Threshold

0.182

SVM

0.275

VAE oracle

0.814

LSTM oracle

0.011 $\rightarrow$ 0.814 on one channel — the central argument in miniature.

Five controls that make the results defensible

1

External reproduction

Reproduces the published
Hundman (2018)
benchmark on SMAP.

Δ 0.006

2

Leakage-free protocol

Channel-level 61 / 20 split,
seed 42; the contaminated
run reported and
withdrawn.

61 / 20

3

Statistical testing

Paired Wilcoxon over 20
held-out channels — low
power reported, not
equivalence claimed.

n = 20

4

Generator fidelity suite

Fréchet and ACF on the
clean generator; MMD, KS,
t-SNE were development
checks.

FD + ACF

5

Negative / validity controls

No-injection false-positive
baseline; the synth-median
“win” diagnosed as an
artefact.

FP 0.011

Every headline number in this talk is reported with the control that qualifies it.